Kernix 0.3.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

09 workloads

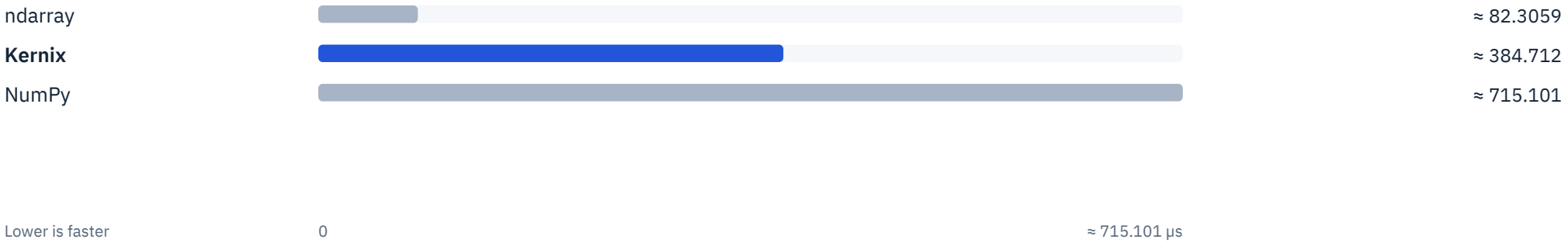
27 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

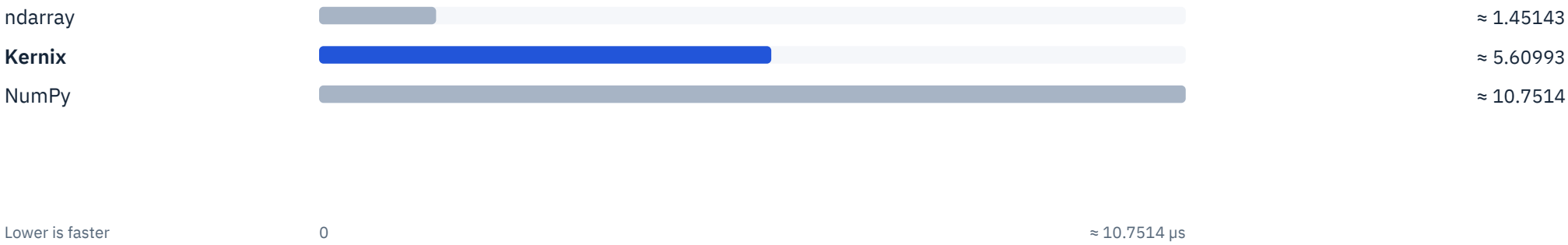
Kernix 0.3.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



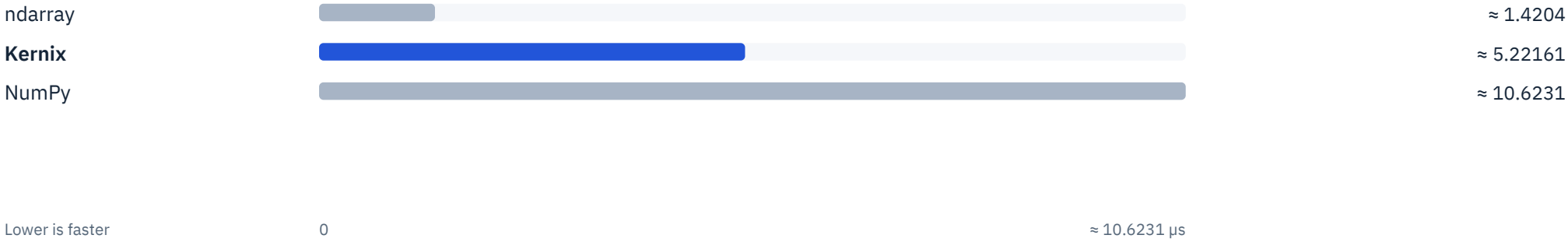
Kernix 0.3.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication · reused output

Batch-mean median

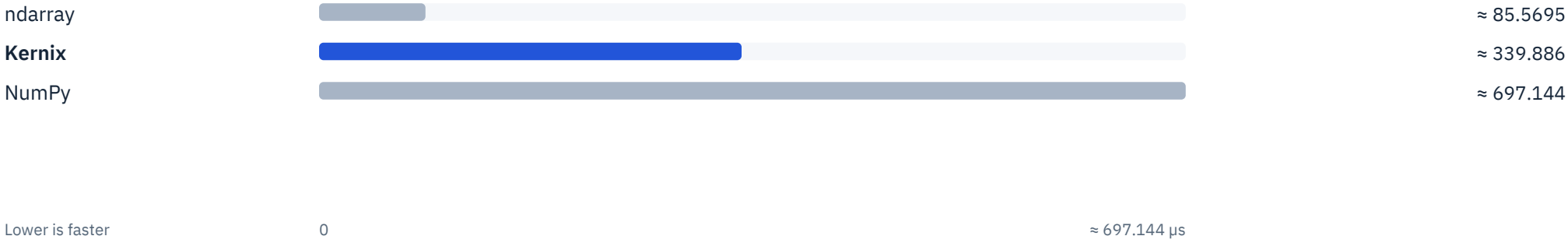
32 × 32 · 32 × 32 · f64 · 1 worker · reused output



Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output



Kernix 0.3.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

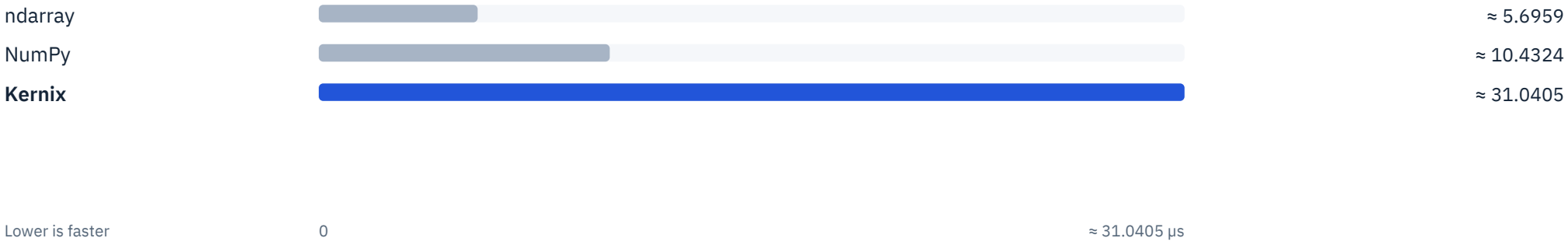
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



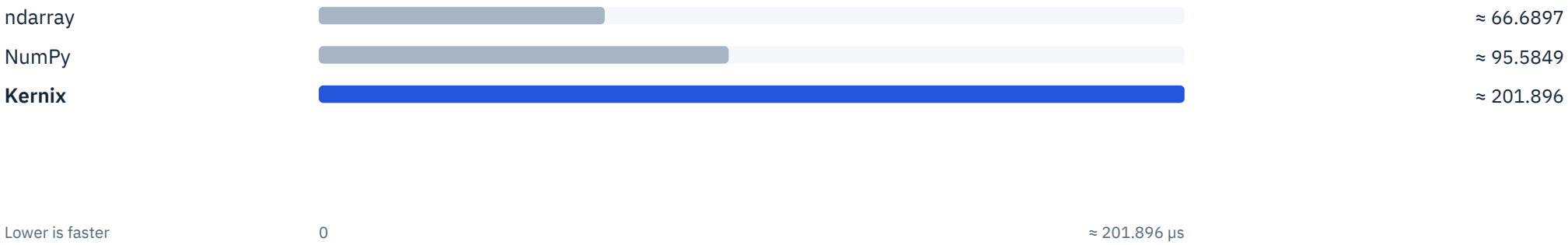
Kernix 0.3.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median



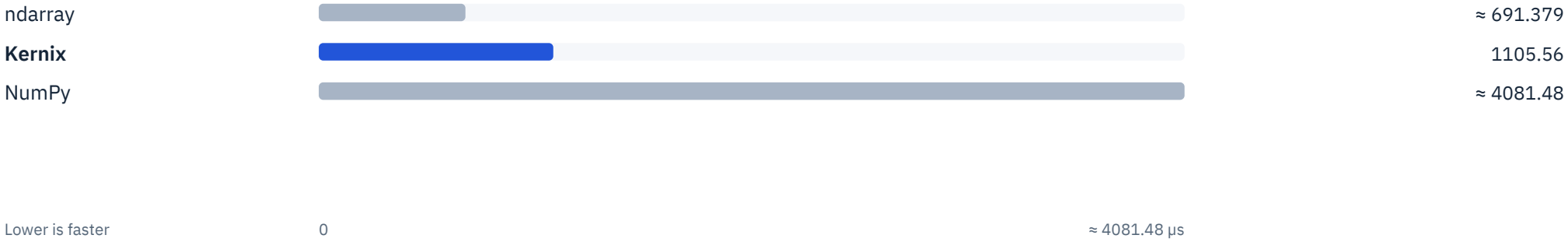
Kernix 0.3.0 / Vision workloads

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

RGB resize & normalization

Batch-mean median

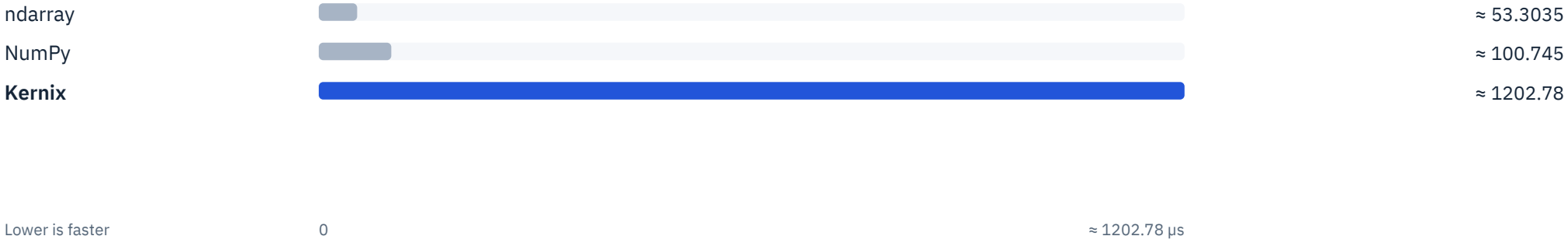
1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 · 1 worker · allocated output · comparator implementations differ



2D convolution

Batch-mean median

1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 · 1 worker · allocated output · comparator implementations differ



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	5609.9288759255105	5.6099288759255105
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1451.4304208998549	1.4514304208998549
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	10751.41143594153	10.75141143594153
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	384712.3846153846	384.7123846153846
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	82305.86513157895	82.30586513157895
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	715100.9142857143	715.1009142857143
Matrix multiplication · reused output / Kernix 32 × 32 · 32 × 32 · f64	5221.606516290727	5.221606516290727
Matrix multiplication · reused output / ndarray 32 × 32 · 32 × 32 · f64	1420.400999943185	1.420400999943185
Matrix multiplication · reused output / NumPy 32 × 32 · 32 × 32 · f64	10623.101954120646	10.623101954120646
Matrix multiplication · reused output / Kernix 128 × 128 · 128 × 128 · f64	339885.6216216216	339.8856216216216

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · reused output / ndarray 128 × 128 · 128 × 128 · f64	85569.47781569966	85.56947781569966
Matrix multiplication · reused output / NumPy 128 × 128 · 128 × 128 · f64	697144.3055555555	697.1443055555555
Elementwise addition / Kernix 65,536 elements · f64	21655.509956709957	21.655509956709957
Elementwise addition / ndarray 65,536 elements · f64	22168.906028368794	22.168906028368794
Elementwise addition / NumPy 65,536 elements · f64	22975.808999081728	22.975808999081728
Sum reduction / Kernix 65,536 elements · f64	31040.481389578163	31.040481389578163
Sum reduction / ndarray 65,536 elements · f64	5695.9009111617315	5.6959009111617315
Sum reduction / NumPy 65,536 elements · f64	10432.436795994994	10.432436795994994
Row softmax / Kernix 256 × 64 · f64	201896.40322580645	201.89640322580645
Row softmax / ndarray 256 × 64 · f64	66689.74933333334	66.68974933333334

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	95584.92366412214	95.58492366412214
RGB resize & normalization / Kernix 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	1105560	1105.56
RGB resize & normalization / ndarray 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	691379.1891891892	691.3791891891892
RGB resize & normalization / NumPy 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	4081478.285714286	4081.478285714286
2D convolution / Kernix 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	1202782.5714285714	1202.7825714285714
2D convolution / ndarray 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	53303.54893617021	53.30354893617021
2D convolution / NumPy 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	100744.53413654618	100.74453413654618